

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Explain, in terms of collision theory, why increasing temperature increases the rate of reaction.

[5 marks]

2

Determine the concentration of a solution made by dissolving 5.85 g of NaCl in 250 cm³ of water.

[5 marks]

$$c = \frac{n}{V}$$

3

Calculate the percentage yield when 8.4 g of product is obtained from a theoretical maximum of 10.0 g.

[4 marks]

4

Explain why ionic compounds conduct electricity when molten but not when solid.

[5 marks]

5

Describe the trend in reactivity down Group 1 and explain it in terms of atomic structure.

[6 marks]

6

Calculate the relative formula mass of calcium carbonate and the mass of 0.25 mol of it.

[4 marks]

7

Explain the difference between a strong acid and a concentrated acid, with an example of each.

[5 marks]

8

Describe a method to prepare a pure, dry sample of a soluble salt from an insoluble base.

[6 marks]
